# Recent Motion History and 24-Hour Surface-Drifter Position Forecasts in the Gulf Stream: Physical Baselines, Learned Corrections and Motion-Conditioned Error

DATA3001 Data Science and Decisions in Practice — Final Report

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## Executive summary

**Question.** Where will a surface drifter in the Gulf Stream be 24 hours from now, and does knowing how it moved over the previous 24 hours improve that prediction beyond simply extrapolating its present velocity? We also ask under which recent motion conditions—speed, acceleration and turning—24-hour forecasts are most difficult, and where trajectory history helps most.

**Data and design.** Using the NOAA Global Drifter Program (GDP) hourly product (Elipot et al., 2016, 2022b), we built 626,679 forecast cases from 517 drifters with forecast origins in the box 80°–60°W, 30°–45°N during July–September 2007–2022. Each case uses 25 hourly observations ($t - 24$ h to $t$) to predict the observed position at $t + 24$ h. Two physical baselines (Constant Velocity and empirical Mean-flow Advection) were compared with five learned model families (Ridge, Elastic Net, Random Forest, Extra Trees and Gradient Boosting), each fitted with *current-state* features only and with additional *history* features. Learned models predict a correction to the Constant Velocity displacement, which is converted to a geographic position by spherical geodesic formulas. Drifters were split once into disjoint training (361), validation (78) and test (78) sets; hyperparameters were chosen on validation data and the frozen models were evaluated once on the 82,546 held-out test forecasts.

**Main findings (held-out test set).**

- History-enhanced Gradient Boosting (GB History) had the lowest error: median 9.23 km and 90th percentile (P90) 28.55 km, against 15.89 km and 40.90 km for Constant Velocity (reductions of 41.9% and 30.2%). Random Forest and Extra Trees with history performed almost identically (medians 9.25 km and 9.33 km); we do not claim a ranking among the three.
- Adding 24-hour history reduced median error in every model family, by 17–18% for the linear models and 29% for the tree ensembles. For Gradient Boosting, the mean paired error reduction was 3.65 km (95% drifter-cluster bootstrap interval 3.14 km to 4.14 km).
- The Mean-flow Advection baseline, built from a training-data climatology, was the least accurate method overall (median 24.28 km).
- Errors grow with recent speed and acceleration but *fall* with turning rate, largely because fast drifters tend to move straight and slow drifters turn more. The benefit of history was positive in all nine speed–turning regimes and largest for fast, strongly turning drifters (7.03 km, 95% interval 5.38 km to 9.17 km).
- The advantage of GB History held in every one of the 16 years. Sea-surface temperature was associated with forecast difficulty, but adding it as a predictor changed validation error by less than 0.2%.

**Why it matters.** Shrinking the typical 24-hour miss from about 16 km to 9 km is the kind of gain that search-and-rescue, pollution-tracking and ocean-monitoring users of drift estimates care about. Equally useful is knowing *in advance* which forecasts are unreliable: recent speed, which is available at forecast time, is strongly associated with larger errors, whereas track curvature is not a warning sign. These applications were not tested here, and other floating objects drift differently from drogued buoys.

**Recommendation.** For the industry partner and other users of the GDP hourly product, a history-enhanced tree-ensemble correction to Constant Velocity is the recommended benchmark for *retrospective* evaluation of 24-hour drifter predictability, and is clearly preferable to either physical baseline. Gradient Boosting is the

practical choice because it was by far the fastest of the three to train. GDP hourly velocities are smoothed using fixes on both sides of each time, and almost half of the forecast origins come from drifters that had lost their drogue. The results are therefore *not* yet evidence of operational real-time forecasting skill. Recommended next steps are evaluation with real-time inputs, drogue-stratified and chronological hold-outs, multiple lead times and, for applications, validation against the drift of the relevant objects.

# 1 Introduction

## 1.1 Motivation and client context

The upper few metres of the ocean carry oil, plastic debris, larvae, life rafts and lost cargo. Operational search-and-rescue and spill-response systems therefore depend on short-range estimates of where surface water, and objects in it, will be in the next hours to days (Breivik and Allen, 2008; van Sebille et al., 2018). Better 24-hour position estimates matter for three kinds of user. *Search and rescue:* planners derive a search area from a last known position and a drift prediction, and the area needed to contain a target grows with the square of the position uncertainty. Objects such as life rafts, however, also respond to wind ("leeway"), so drifter skill transfers to them only in part (Breivik and Allen, 2008; Breivik et al., 2011). *Pollution and debris tracking:* short-range transport estimates help decide where to monitor and intervene. Oil and floating debris, however, experience windage, wave-induced Stokes drift and other processes that a drogued drifter is designed to avoid (van Sebille et al., 2020). *Oceanographic monitoring:* users of the GDP record need to know when a drifter's recent track is a reliable guide to its near future, and when it is not. This study evaluates none of these applications directly; it quantifies retrospective 24-hour forecast skill for GDP drifters, which is a necessary first step for all three.

Satellite-tracked surface drifters provide the most direct observational record of this motion. The NOAA Global Drifter Program (GDP) maintains an array of drogued buoys that follow near-surface currents. Their tracks underpin global estimates of surface circulation and are a reference for evaluating ocean models and calibrating satellite products (Lumpkin and Johnson, 2013; Centurioni et al., 2019).

This project uses the hourly GDP product (Elipot et al., 2016) and its companion hourly sea-surface temperature (SST) dataset (Elipot et al., 2022a), supplied as the course dataset. The hourly product interpolates irregular Argos and GPS fixes onto a uniform one-hour grid and provides positions, velocities and uncertainty estimates. This resolution is what makes a 24-hour forecasting study with hourly inputs feasible. A question of direct interest to data providers and users is therefore how much short-range predictability can be extracted from the product's own kinematic information, and under which motion conditions simple extrapolation breaks down.

We study the Gulf Stream, a narrow, fast western boundary current whose path downstream of Cape Hatteras meanders and has become less stable in recent decades (Andres, 2016). Strong spatial gradients and rapid changes of direction make it a demanding test region. A drifter moving at $1\,\mathrm{m\,s^{-1}}$ travels about 86 km in 24 hours, so a small error in the assumed velocity quickly becomes a large position error.

## 1.2 Related work

Data-driven forecasting of drifter motion has clear precedents. Aksamit et al. (2020) combined a recurrent neural network trained on drifter motion history with a reduced Maxey–Riley model and found improvements over the physical model alone for Gulf of Mexico deployments. Grossi et al. (2025) predicted Gulf of Mexico drifter trajectories over 24-hour and five-day windows. They reported that fully connected networks using velocity histories did not outperform autoregressive baselines, whereas a model sharing information between drifters performed better in many cases: added flexibility does not guarantee skill. Lin et al. (2026) benchmarked kinematic and machine-learning models for ten-minute forecasts of a single buoy under explicit leakage control. Constant velocity performed best overall, and trajectory-geometry-dependent gains from learned corrections were limited. Together these studies suggest that (i) strong kinematic baselines are essential, (ii) evaluation must prevent leakage between training and test trajectories, and (iii) aggregate scores can conceal condition-specific behaviour. Our study combines these elements at a 24-hour horizon on many held-out drifters. We make no claim to introduce a new forecasting algorithm or to be the first to use motion

history.

### 1.3 Research question, objectives and scope

The primary research question, fixed in the project proposal, is:

> *How do 24-hour position forecast errors vary with a drifter's recent speed, acceleration and turning, and under which motion conditions does trajectory history improve prediction beyond current-state information and physical baselines?*

The objectives were to (O1) construct comparable 24-hour forecasts from Constant Velocity, Mean-flow Advection and learned residual-correction models; (O2) quantify how median and upper-tail errors vary with pre-forecast motion characteristics; (O3) compare each learned model with and without history features to identify where history helps; and (O4) report the results reproducibly, with uncertainty that respects the dependence between forecasts from the same drifter. We extended the original plan with three further model families, spatial, temporal and SST analyses, and an SST feature ablation.

The scope is deliberately narrow: one region, one season (July–September), one forecast horizon (24 h) and one history length (24 h). Forecast "difficulty" is defined relative to the tested methods and data, not as an intrinsic limit on ocean predictability, and all regime comparisons are associations rather than causal effects.

Sections 2 to 4 describe the data, forecasting framework and experimental design. Sections 5 to 7 present the results, and Sections 8 and 9 discuss limitations, recommendations and conclusions. Section 10 reflects on the use of AI tools.

## 2 Data and forecast-sample construction

### 2.1 The GDP hourly product

We use version 2.01 of the GDP hourly dataset (Elipot et al., 2022b). Each record gives a drifter identifier, UTC time, longitude $\lambda$, latitude $\phi$, eastward and northward velocity $(u, v)$ in m s$^{-1}$, SST, and a drogue-status flag. Two properties of the product matter for interpretation. First, positions and velocities on the hourly grid are *estimates* obtained by fitting irregular satellite fixes, not direct hourly measurements. The fit at time $t$ draws on fixes on either side of $t$ (Elipot et al., 2016), so the velocity at a forecast origin carries a small amount of information from the following hours. Second, most drifters eventually lose their drogue, after which wind slip makes the recorded velocity no longer representative of the 15 m current. The final cohort was not filtered on drogue status.

### 2.2 Cohort definition

Forecast origins were restricted to the box 80°–60°W, 30°–45°N and to July–September of 2007–2022 (Fig. 1a). Historical inputs and targets were allowed to fall outside the box or season; for example, a 1 July origin may use 30 June history. All observations were retrieved from the complete trajectories of drifters that entered the region (20,216,717 hourly records).

A forecast case at origin $t$ requires (i) all 25 hourly records from $t - 24$ h to $t$ with valid position and velocity, and (ii) an observed position at exactly $t + 24$ h. Records were matched by drifter identifier and timestamp rather than by row offset, so gaps in a track cannot silently shift the horizon. Of 632,770 eligible origins from 519 drifters, 626,679 (99.04%) from 517 drifters satisfied both requirements (Table 1). Every method is evaluated on exactly the same cases, including the baselines that do not use history.

### 2.3 Data characteristics relevant to the analysis

Recent motion in the cohort is strongly right-skewed. The 24-hour mean speed has median 0.31 m s$^{-1}$ and mean 0.41 m s$^{-1}$, with a maximum of 2.60 m s$^{-1}$. The mean turning rate has median 9.4 ° h$^{-1}$. Sampling is uneven in space and time. Origins concentrate in the southern and central parts of the box, whereas the northern Gulf Stream extension is sparsely and patchily sampled (Fig. 1a). Annual sample sizes vary by a factor of almost three (Fig. 1b). Origin-time SST was available for 98.4%, 97.0% and 94.3% of training, validation and test cases. Finally, 48.7% of origins come from drifters flagged as undrogued (Table 1). This

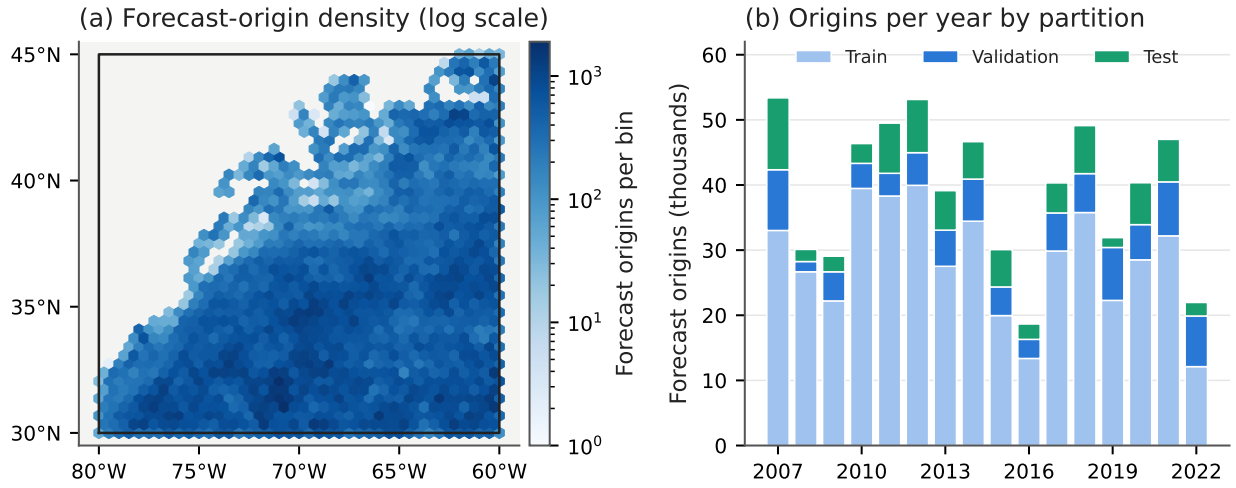



**Figure 1:** Study region and sample coverage. (a) Density of the 626,679 forecast origins (logarithmic colour scale); the black rectangle is the origin-selection box. Land and unsampled areas are blank; no coastline is drawn. (b) Number of forecast origins per year by drifter-disjoint partition. Between 25 and 54 distinct drifters contribute in each year. Generated from `modelling_dataset_final.parquet` (Appendix A.2).

**Table 1:** Forecast cohort and drifter-disjoint partitions. The split assigns whole drifters, so sample shares differ from the nominal 70/15/15 drifter shares. "Undrogued" is the share of forecast origins whose drogue-status flag is false at $t$, computed for this report as a descriptive check (not used by any model).

| Partition | Forecast cases | Share of cases | Drifters | Undrogued origins |
|---|---|---|---|---|
| Training | 455,662 | 72.7% | 361 | 48.2% |
| Validation | 88,471 | 14.1% | 78 | 53.6% |
| Test | 82,546 | 13.2% | 78 | 46.0% |
| Total | 626,679 | 100% | 517 | 48.7% |

proportion was identified only during report preparation; Section 8.3 discusses its consequences.

# 3 Forecasting framework and features

## 3.1 Notation and forecast target

For drifter $i$ at forecast origin $t$, let $\mathbf{p}_t = (\lambda_t, \phi_t)$ be the position, $\mathbf{v}_t = (u_t, v_t)$ the eastward and northward velocity, and $\tau = 86{,}400\,\text{s}$ the 24-hour horizon. The input window is $\mathcal{W}_t = \{(\mathbf{p}_{t-k}, \mathbf{v}_{t-k}) : k = 0, \ldots, 24\}$ and the target is the observed position $\mathbf{p}_{t+\tau}$ (Fig. 2). Every forecast is a *direct* 24-hour endpoint prediction. No model recursively generates intermediate hourly positions; the stepwise mean-flow baseline integrates only its own climatological field.

Displacements are expressed as east–north vectors $\mathbf{d} = (d_E, d_N)$ in metres on a sphere of radius $R = 6{,}371\,\text{km}$. The *forward geodesic* map $\mathcal{G}(\mathbf{p}, \mathbf{d})$ moves a distance $s = \|\mathbf{d}\|$ along a great circle with initial bearing $\beta = \operatorname{atan2}(d_E, d_N)$:

$$\phi' = \arcsin\big(\sin\phi\cos\delta + \cos\phi\sin\delta\cos\beta\big), \qquad \lambda' = \lambda + \operatorname{atan2}\big(\sin\beta\sin\delta\cos\phi,\ \cos\delta - \sin\phi\sin\phi'\big), \quad (1)$$

with $\delta = s/R$. The *inverse geodesic* $\mathcal{G}^{-1}(\mathbf{p}, \mathbf{p}')$ returns $\mathbf{d} = R\,\sigma\,(\sin\beta, \cos\beta)$, where $\sigma$ is the central angle and $\beta$ the initial bearing from $\mathbf{p}$ to $\mathbf{p}'$. Forecast error is the great-circle (haversine) distance

$$E = D(\hat{\mathbf{p}}, \mathbf{p}_{t+\tau}) = 2R \arcsin\sqrt{\sin^2\tfrac{\Delta\phi}{2} + \cos\hat{\phi}\,\cos\phi_{t+\tau}\,\sin^2\tfrac{\Delta\lambda}{2}}. \quad (2)$$

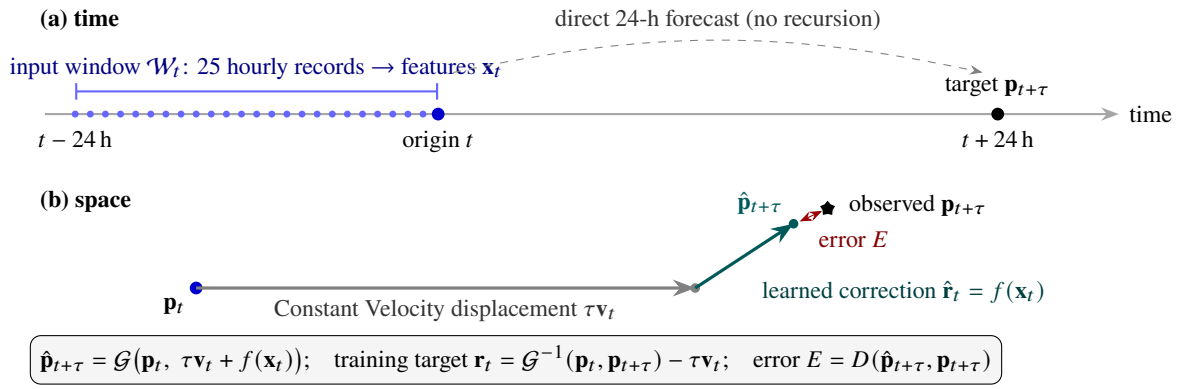



**Figure 2:** Forecast set-up. A forecast issued at origin $t$ uses only the 25 hourly records up to and including $t$. Learned models predict the residual between the observed 24-hour displacement and the Constant Velocity displacement. The predicted displacement is mapped to a position with the forward geodesic (1).

## 3.2 Physical baselines

**Constant Velocity (CV).** The current velocity is extrapolated for 24 hours: $\hat{\mathbf{p}}^{\text{CV}}_{t+\tau} = \mathcal{G}(\mathbf{p}_t, \tau\mathbf{v}_t)$.

**Mean-flow Advection.** A climatological velocity field $\bar{\mathbf{v}}(c)$ was estimated on a $1°\times1°$ grid as the mean origin velocity of all *training* forecasts in cell $c$ (224 cells). A particle is advected in 24 hourly steps,

$$\mathbf{p}^{(k+1)} = \mathcal{G}\big(\mathbf{p}^{(k)},\ \Delta t\,\bar{\mathbf{v}}(c(\mathbf{p}^{(k)}))\big), \quad k = 0, \ldots, 23, \quad \Delta t = 3{,}600\,\text{s}, \quad \mathbf{p}^{(0)} = \mathbf{p}_t, \quad (3)$$

using the drifter's own origin velocity whenever the particle enters a cell with no training data (0.73% of hourly steps on validation data). This stepwise version replaced an initial single-step version (one cell mean held for 24 h) after it reduced validation median error slightly, from 22.66 km to 22.37 km.

## 3.3 Residual-correction learning

Learned models do not predict positions directly. They estimate a correction to Constant Velocity:

$$\mathbf{r}_t = \mathcal{G}^{-1}(\mathbf{p}_t, \mathbf{p}_{t+\tau}) - \tau\mathbf{v}_t, \qquad \hat{\mathbf{r}}_t = f(\mathbf{x}_t), \qquad \hat{\mathbf{p}}_{t+\tau} = \mathcal{G}\big(\mathbf{p}_t,\ \tau\mathbf{v}_t + \hat{\mathbf{r}}_t\big). \quad (4)$$

Here $f : \mathbb{R}^p \to \mathbb{R}^2$ is fitted by least squares on the two residual components. This formulation makes CV the natural "zero-correction" reference, so a learned model only needs to describe *departures* from persistence of velocity. Note that models are fitted with a squared-error loss in displacement space but evaluated and tuned on great-circle errors (Section 4).

**Geometric consistency.** In a first implementation, observed displacements were computed in a local Cartesian approximation, $d_E = R\cos\phi_t\,\Delta\lambda$ and $d_N = R\,\Delta\phi$, while predictions were mapped back with the spherical forward geodesic. To test this mismatch, we reconstructed each observed validation target from its own local-Cartesian displacement. The round-trip error had median 0.021 km and 99th percentile 1.54 km (maximum 2.98 km), and 1,961 validation cases (2.2%) exceeded 1 km. The discrepancy is small relative to forecast errors, but approximation error of this kind is expected to grow with displacement and latitude, and long displacements are exactly where the motion analysis focuses. Targets were therefore redefined with the inverse geodesic in (4), after which the round-trip error is at machine precision (maximum $5 \times 10{-}9$ m), and all learned models were refitted. All reported results use the refitted models.

### 3.4 Features

Table 2 lists the 25 predictors. The *current-state* set (9 features) describes only the origin; the *history* set adds 16 summaries of the input window. With $s_k = \|\mathbf{v}_{t-k}\|$ and $\Delta t = 3{,}600$ s, the three motion descriptors used later to define regimes are

$$\bar{s}_t = \frac{1}{25}\sum_{k=0}^{24} s_k, \qquad \bar{a}_t = \frac{1}{24}\sum_{k=0}^{23} \frac{\|\mathbf{v}_{t-k} - \mathbf{v}_{t-k-1}\|}{\Delta t}, \qquad \bar{\omega}_t = \frac{1}{|\mathcal{K}_t|}\sum_{k\in\mathcal{K}_t} \left|\,\mathrm{wrap}(\theta_{t-k} - \theta_{t-k-1})\right|, \tag{5}$$

where $\theta = \mathrm{atan2}(v, u)$ is the heading, wrap maps angles to $(-180^\circ, 180^\circ]$, and $\mathcal{K}_t$ contains the hourly steps at which both speeds are at least $0.05\ \mathrm{m\,s{-}1}$, because heading is unreliable near zero speed. The mean turning rate $\bar{\omega}_t$ is in degrees per hour and is undefined if $\mathcal{K}_t$ is empty (471 of 626,679 cases). All features use only information available at or before $t$.

**Table 2:** Predictors. Current-only models use the 9 current-state features; history-enhanced models use all 25.

| Group | Features |
|---|---|
| Current state (9) | $\lambda_t$, $\phi_t$, $u_t$, $v_t$, speed $s_0$, heading $\sin\theta_t$ and $\cos\theta_t$, day of year as sin and cos |
| History: speed and velocity (6) | mean and s.d. of speed; mean and s.d. of $u$ and of $v$ over the 25 records |
| History: acceleration (3) | mean and maximum acceleration magnitude; mean absolute speed change per second |
| History: turning (3) | mean and maximum absolute turning rate; fraction of valid turning steps $\lvert\mathcal{K}_t\rvert/24$ |
| History: trend and displacement (4) | least-squares slopes of $u$ and $v$ over the window; net change in longitude and latitude from $t - 24$ h to $t$ |

## 4 Models and experimental design

### 4.1 Learned model families

Five families were fitted to the residual target (4) using scikit-learn (Pedregosa et al., 2011), each in a *Current* (9 features) and a *History* (25 features) variant. The paired variants share the estimator, preprocessing, tuning grid and data, so their difference isolates the contribution of history features.

- **Ridge regression** (Hoerl and Kennard, 1970) and **Elastic Net** (Zou and Hastie, 2005): linear corrections with $\ell_2$, and combined $\ell_1/\ell_2$, penalties. Missing turning summaries were median-imputed (training medians) with missing-value indicators, and features were standardised. The multi-task Elastic Net applies a joint penalty to the eastward and northward coefficients.
- **Random Forest** (Breiman, 2001) and **Extra Trees** (Geurts et al., 2006): bagging-type tree ensembles that capture nonlinearity and interactions, with 80% of features considered per split and median imputation. Extra Trees used randomised split thresholds without bootstrap resampling.
- **Gradient Boosting** (Friedman, 2001): histogram-based boosted trees (one model per displacement component), $\ell_2$ regularisation 1.0, minimum leaf size 50, with native missing-value handling. The notebook did not override scikit-learn's default `early_stopping="auto"`, which for more than 10,000 samples stops boosting when the loss on an internal validation set stops improving. That internal set is a *random row-level* 10% of the training forecasts, not a drifter-grouped split. Because consecutive fore-

casts from the same drifter overlap almost completely, internal validation rows have near-duplicates in the fitting rows, so the internal loss is optimistic and may allow more boosting iterations than a drifter-grouped split would. The iteration counts in Table 3 are therefore upper bounds. This affects only the number of trees fitted on *training* drifters; the external validation and test drifters were never part of this internal split, so the held-out evaluation is not contaminated.

**Table 3:** Hyperparameter grids and validation-selected configurations (criterion: minimum validation median great-circle error). Tree configurations are (number of trees, maximum depth, minimum leaf size) and boosting configurations are (maximum iterations, maximum leaf nodes, learning rate).

| Family | Candidate grid | Selected: Current | Selected: History |
|---|---|---|---|
| Ridge | $\alpha \in \{0.1, 1, 10, 100, 1000\}$ | $\alpha = 1000$ | $\alpha = 100$ |
| Elastic Net | $\alpha \in \{0.01, 0.1, 1, 10\} \times \ell_1$ ratio $\in \{0.2, 0.5, 0.8\}$ | $\alpha = 1$, ratio 0.5 | $\alpha = 0.01$, ratio 0.8 |
| Random Forest | (100,16,20), (150,20,20), (150,24,10) | (100,16,20) | (150,24,10) |
| Extra Trees | (150,16,20), (150,20,20), (150,24,10) | (150,24,10) | (150,24,10) |
| Gradient Boosting | (150,15,0.08), (200,31,0.06), (250,31,0.08) | (250,31,0.08) | (250,31,0.08) |

## 4.2 Partitioning, tuning and final evaluation

Overlapping 25-hour windows from the same drifter are highly correlated. A random split of forecast *cases* would place near-duplicate cases in training and test and overstate skill. We therefore used a **single drifter-disjoint split**, assigning each drifter to exactly one partition: 15% of drifters to test, then 15/85 of the remainder to validation (`GroupShuffleSplit`, seeds 42 and 43). This is a hold-out design, *not* $K$-fold cross-validation. Every reported number comes from one realisation of the split, and the sensitivity of the results to which drifters fall in the test set has not been measured. The mean-flow field, imputation medians, scalers and motion-regime thresholds were all estimated from training data only.

Hyperparameters were selected by validation median error from the small grids in Table 3. Median error was chosen as a robust criterion for a heavy-tailed outcome; RMSE would have emphasised rare large errors. The history variants of all three tree families selected the most complex configuration in their grid, so additional capacity might improve them further. After selection, configurations were frozen and *all twelve* methods were evaluated once on the untouched test drifters, with no further tuning. The motion, spatial, temporal and SST analyses (Sections 6 and 7) reuse these frozen test predictions and are best regarded as exploratory diagnostics. The SST feature experiment was deliberately evaluated on the validation set, because the test set had already been examined.

## 4.3 Metrics and forecast interpretation

For each method we report the mean, median, 90th percentile (P90) and root-mean-square (RMSE) of the great-circle error $E$ (2) over identical cases. The median describes a typical forecast, while P90 and RMSE describe the upper tail that matters most for applications such as search planning. Each forecast is a *rolling* 24-hour forecast: it is re-initialised at every origin from the *observed* state and history. Plots that join successive forecast endpoints (Section 5.2) therefore show a sequence of independent 24-hour predictions, *not* an autonomous trajectory in which predictions are fed back as inputs. Skill in this setting says nothing directly about free-running multi-step forecasting.

## 4.4 Motion regimes and uncertainty quantification

Test forecasts were stratified into Low, Medium and High regimes of $\bar{s}_t$, $\bar{a}_t$ and $\bar{\omega}_t$ (5). The regimes are bounded by the 33.3rd and 66.7th percentiles of the *training* distribution: $0.239\,\mathrm{m\,s^{-1}}$ and $0.399\,\mathrm{m\,s^{-1}}$ for speed, $1.55 \times 10^{-5}\,\mathrm{m\,s^{-2}}$ and $2.33 \times 10^{-5}\,\mathrm{m\,s^{-2}}$ for acceleration and $6.70\,^\circ\mathrm{h^{-1}}$ and $13.98\,^\circ\mathrm{h^{-1}}$ for turning. Thirteen test forecasts from one drifter with undefined turning were kept as a separate category and excluded from turning-based summaries.

Forecasts from the same drifter are dependent, because consecutive origins share 24 of their 25 input records and their targets are only an hour apart. Treating the 82,546 test cases as independent would grossly understate uncertainty. We therefore used a **drifter-cluster bootstrap** (Efron and Tibshirani, 1993; Cameron et al.,

2008). For regime $k$, the paired benefit of history, $g_{ij} = E_{ij}^{\mathrm{GBC}} - E_{ij}^{\mathrm{GBH}}$ for forecast $j$ of drifter $i$, is summarised by the ratio estimator

$$\hat{G}_k = \frac{\sum_i S_{ik}}{\sum_i N_{ik}}, \qquad S_{ik} = \sum_{j \in k} g_{ij}, \quad N_{ik} = \#\{j \in k\}, \tag{6}$$

and each of $B = 2000$ replicates resamples the 78 test drifters with replacement, keeping all of their forecasts. Percentile intervals (2.5%, 97.5%) are reported. This accounts for within-drifter dependence and unequal track lengths. It does not account for dependence between different drifters sampling the same eddy, nor for the selection of comparisons after inspecting the data.

# 5 Overall forecast performance

## 5.1 Validation and held-out test results

Table 4 reports validation medians, which guided model selection, alongside the final test metrics, which are the evidence for our conclusions. Validation and test give the same ranking of methods. Errors are uniformly larger on the test drifters, by 3.1–10.5% in median (CV 15.04 km on validation against 15.89 km on test, +5.6%; GB History 8.51 km against 9.23 km, +8.5%). The untuned CV baseline shifts as well, so part of the difference reflects the composition of the two sets of 78 drifters; the test drifters are, for example, faster on average (Section 6). However, the History variants shift more in relative terms (8.2–10.5%) than CV, and validation estimates of models chosen as the best of several configurations are expected to be optimistic. Some validation-set optimism from hyperparameter selection therefore cannot be ruled out. For this reason, all conclusions rest on the test results, on which no choice was made.

**Table 4:** Validation median and held-out test error (km) for all methods on identical cases. The last column is the reduction in test median relative to Constant Velocity (negative means worse). Test set: 82,546 forecasts from 78 drifters. Best values in bold.

| | Validation | Test | | | | Median vs CV |
|---|---|---|---|---|---|---|
| Method | median | mean | median | P90 | RMSE | (%) |
| Constant Velocity | 15.04 | 20.55 | 15.89 | 40.90 | 26.77 | – |
| Mean-flow Advection | 22.37 | 32.38 | 24.28 | 68.80 | 42.05 | −52.8 |
| Ridge Current | 13.24 | 18.49 | 14.14 | 37.70 | 24.00 | 11.0 |
| Ridge History | 10.48 | 16.15 | 11.58 | 33.56 | 22.20 | 27.1 |
| Elastic Net Current | 13.11 | 18.29 | 13.95 | 36.94 | 23.90 | 12.2 |
| Elastic Net History | 10.49 | 16.16 | 11.60 | 33.59 | 22.23 | 27.0 |
| RF Current | 12.62 | 17.00 | 13.05 | 34.37 | 22.07 | 17.8 |
| RF History | 8.54 | 13.33 | 9.25 | 28.81 | 18.78 | 41.7 |
| ET Current | 12.61 | 17.12 | 13.16 | 34.65 | 22.20 | 17.2 |
| ET History | 8.63 | 13.37 | 9.33 | 28.91 | **18.75** | 41.2 |
| GB Current | 12.66 | 16.89 | 13.05 | 34.05 | 21.83 | 17.9 |
| GB History | **8.51** | **13.23** | **9.23** | **28.55** | 18.60 | **41.9** |

**Sensitivity to tuning.** Validation results were insensitive to the Ridge penalty (medians within 0.02 km across $\alpha$ for both variants) and, for the Current tree models, to tree complexity (RF, ET and GB Current medians within 12.61 km to 12.71 km across configurations). The History tree models, by contrast, improved steadily with capacity: GB History from 9.11 to 8.51 km, RF History from 8.83 to 8.54 km and ET History from 9.30 to 8.63 km across their three configurations. History information is therefore what gives the extra model capacity something to learn, and the History models might improve further beyond the grids tested.

**Learned corrections outperform both physical baselines.** Every learned model improves on Constant Velocity in all four metrics. GB History reduces the test median by 41.9% (15.89→9.23 km), the mean by 35.6%, P90 by 30.2% and RMSE by 30.5% (Fig. 3a). A 24-hour displacement correction learned from the drifters' own kinematics therefore removes a substantial part of the error made by assuming that velocity persists.

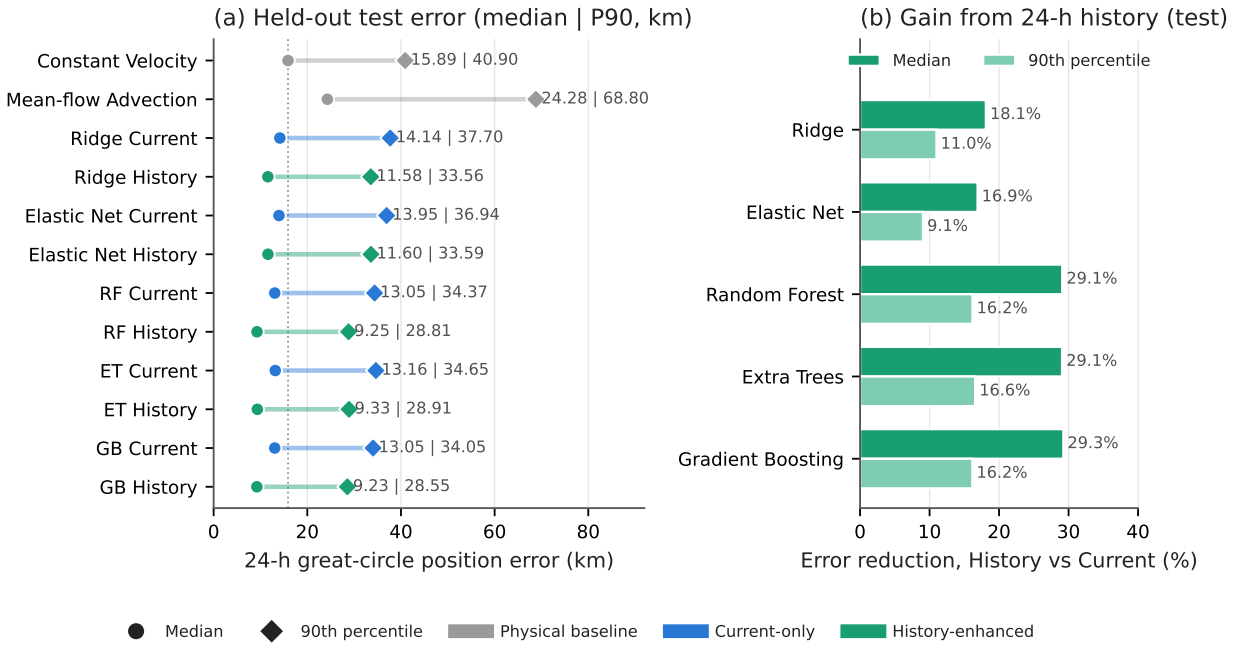



**Figure 3:** Held-out test performance. (a) Median (circle) and P90 (diamond) error for all twelve methods; the dotted line marks the CV median. (b) Relative reduction in median and P90 error from adding history features within each model family. Generated from `models/final_test_comparison.csv`; no model was re-evaluated.

**The empirical mean flow is a weak baseline.** Mean-flow Advection is the least accurate method (median 24.28 km, P90 68.80 km), roughly 50% worse than Constant Velocity. A stationary, seasonally averaged field on a 1° grid smooths away the meanders and eddies that dominate a particular drifter's next 24 hours. Averaging across a narrow jet is also likely to dilute peak speeds, although we did not quantify this. The current velocity, by contrast, already contains this synoptic information. Section 6 shows that the mean flow is nonetheless competitive in one specific regime.

**History features matter more than model flexibility.** Within every family, the History variant beats its Current counterpart (Fig. 3b). The median reduction is 18.1% for Ridge, 16.9% for Elastic Net, and 29.1–29.3% for Random Forest, Extra Trees and Gradient Boosting. Flexibility alone helps less: among Current models, moving from Ridge to Gradient Boosting lowers the median by only 1.09 km (14.14→13.05 km), whereas adding history to Gradient Boosting lowers it by 3.82 km. The two effects interact. In relative terms, history helps the tree ensembles roughly 1.6–1.7 times as much as it helps the linear models, which suggests that the useful information in the history summaries enters partly through nonlinear combinations, for example a velocity trend whose implication depends on current speed. For Gradient Boosting, the overall mean paired reduction from history is 3.65 km with a 95% drifter-cluster bootstrap interval of 3.14 km to 4.14 km (Section 4.4). The benefit is therefore not an artefact of a few long tracks. It is also not universal: in a report-time check of the saved predictions, GB History beat GB Current in 71.4% of individual test forecasts.

**Typical errors improve more than tail errors.** The relative gain in P90 is consistently smaller than in the median: 16.2% against 29.3% for history within Gradient Boosting, and 30.2% against 41.9% for GB History against CV. Even the best model's P90 (28.5 km) is three times its median, so roughly one forecast in ten remains in error by more than 28 km. For applications driven by worst cases, the residual tail is as important as the improvement in the centre.

**The three tree ensembles are practically equivalent.** RF, ET and GB History differ by at most 0.10 km in test median and 0.36 km in P90, and these differences have not been assessed against drifter-level sampling variability. We therefore do not rank them. Gradient Boosting is preferred on practical grounds: on the same machine, the selected configuration trained in about 6 s, compared with about 41 s for Extra Trees and 173 s for Random Forest (validation runs, notebook cells 31–34), and its saved model is roughly 2 MB, compared

with hundreds of megabytes for the bagged ensembles.

## 5.2 Trajectory-level behaviour

Figure 4 joins successive rolling 24-hour forecast endpoints for the three test drifters with the most test forecasts, over the first seven days of each drifter's longest continuous segment. The drifters were chosen by data coverage, not accuracy, but three drifters are illustrations, not a representative sample.

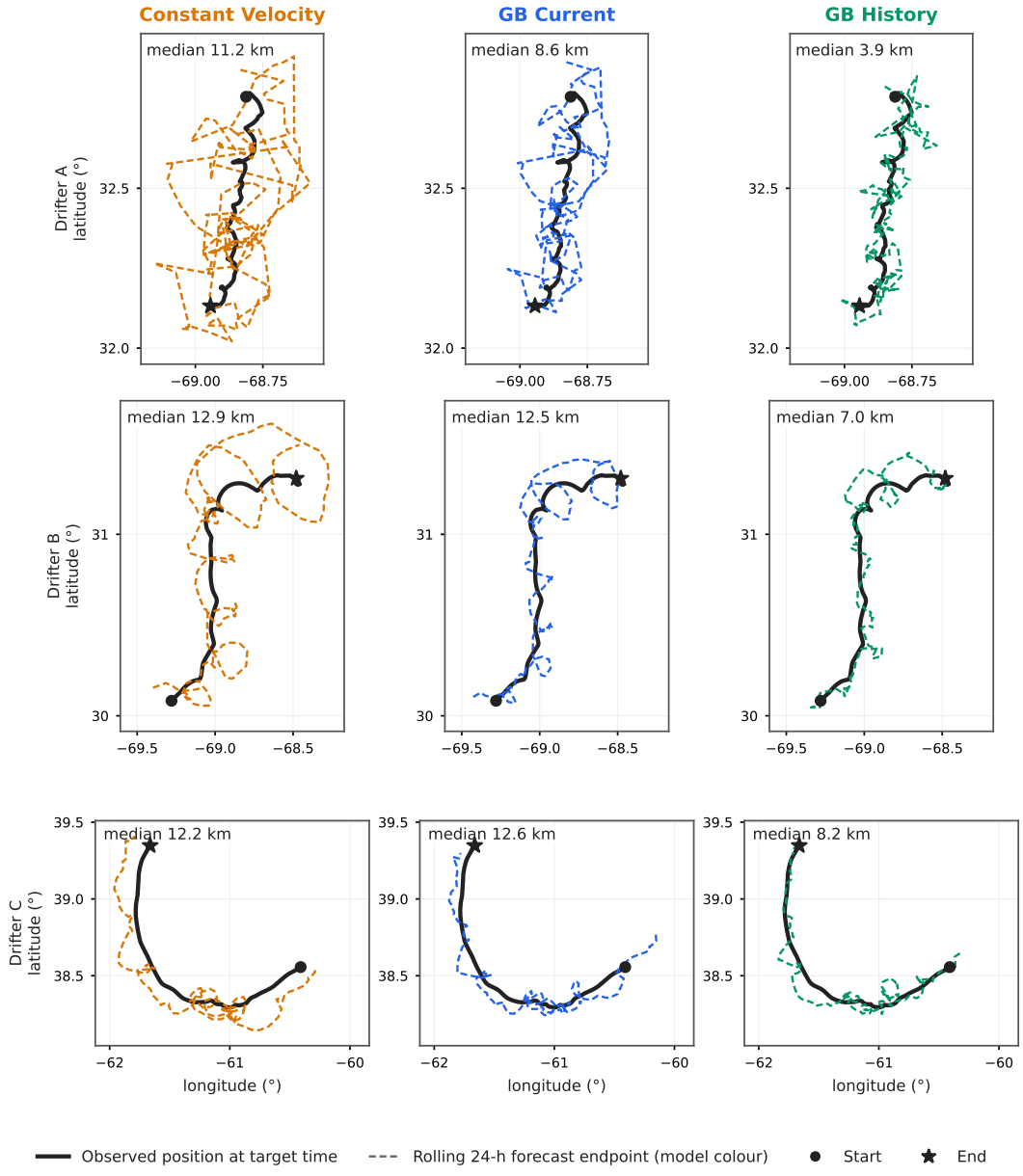



**Figure 4:** Observed positions at target times (black) and rolling 24-hour forecast endpoints (dashed) for CV, GB Current and GB History, for three held-out drifters (A: 139080, B: 300234063940890, C: 300234064909860) over seven days (circle: start; star: end). Each dashed vertex is an independent forecast issued 24 h earlier from the observed state, not a simulated trajectory. In-panel values are window-specific median errors. Axes are fixed per drifter across this figure and Figs. 12 and 13. Re-plotted for legibility from saved test predictions with the selection rules of notebook cell 44, reproducing its panel medians.

Three patterns recur. First, CV endpoints scatter widely and form loops around the observed path. This pattern is consistent with short-period fluctuations in the instantaneous velocity being extrapolated over a full day, although we did not test that mechanism. Second, GB Current contracts the scatter but keeps much of its structure, since it sees only the same instantaneous state. Third, GB History follows the observed path most closely, including through the bend of the third drifter (window medians 3.9, 7.0 and 8.2 km, against 11.2,

12.9 and 12.2 km for CV). It still shows local jitter between consecutive forecasts: accurate endpoints do not imply a smooth sequence of forecasts. The family comparisons in Figs. 12 and 13 (Appendix A.3) show the same picture: Ridge and Elastic Net produce nearly identical tracks, and RF, ET and GB History are visually indistinguishable, matching Table 4.

Figure 5 shows four single forecasts chosen by reproducible rules. The two history-related cases come from the same drifter, which illustrates that the benefit of history varies along one trajectory. In the "difficult" case (the 95th percentile of GB History error), every model extrapolates roughly along the recent north-eastward direction of travel and misses the subsequent curve in the observed path. In the "history helps" case, GB History reduces the error relative to GB Current but still misses by 55.5 km: the drifter travels much farther west-southwest than any model predicts. In the "history hurts" case, the history model over-extrapolates northward while GB Current lands closer. These examples make concrete why the error distribution has a heavy upper tail: some 24-hour futures are simply not predictable from the preceding day of kinematics.

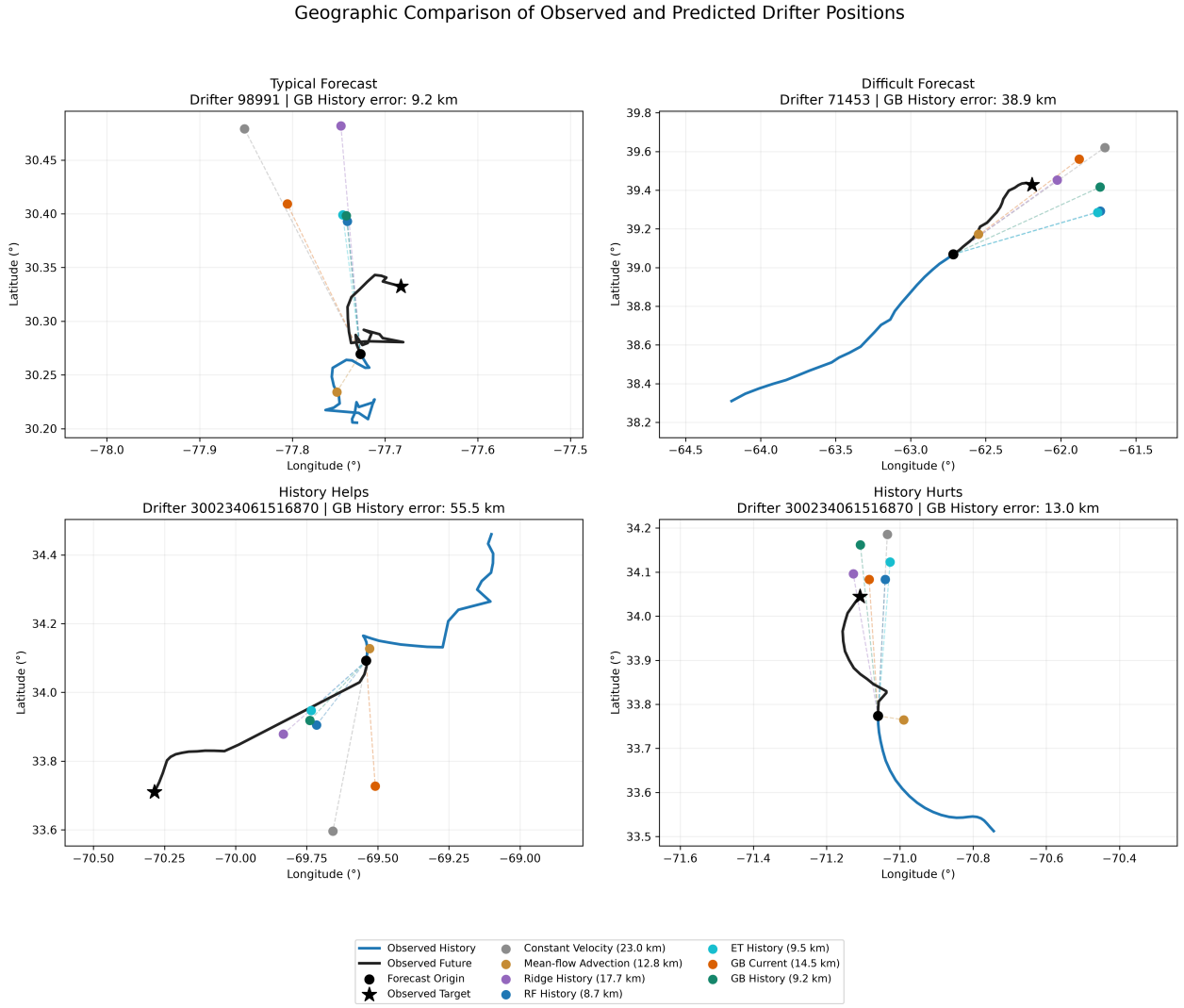



**Figure 5:** Single-forecast case studies on test drifters: typical (GB History error closest to its median), difficult (closest to its 95th percentile), and history helps or hurts (5th and 95th percentiles of the paired gain $E^{\mathrm{GBC}} - E^{\mathrm{GBH}}$). Blue: observed 24-h history; black: observed future; dots: predicted endpoints. *The error values in the shared legend refer to the "Typical" panel only*; panel titles give GB History errors. Source: notebook cell 41.

# 6 Motion-conditioned forecast accuracy

## 6.1 Single-descriptor regimes

Table 5 and Fig. 6 stratify test errors by training-defined tertiles of 24-hour mean speed, acceleration magnitude and turning rate. The test drifters are faster than the training drifters: 41% of test forecasts fall in the training-defined High-speed tertile rather than 33%. This compositional difference is consistent with the higher overall test errors in Table 4. Each regime contains forecasts from 62–77 drifters, so none rests on a handful of tracks.

**Table 5:** Test error (km) by motion regime (training-tertile thresholds in Section 4.4). MF: Mean-flow Advection. The last column is the reduction in median error of GB History relative to CV. Source: `models/motion_conditioned_test_errors.csv` (notebook cell 51).

| | | | | CV | | MF | GBC | GB History | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Descriptor | Regime | Cases | Drifters | median | P90 | median | median | median | P90 | vs CV |
| Speed | Low | 24,510 | 62 | 10.90 | 21.40 | 12.34 | 8.62 | 5.55 | 14.02 | 49.1% |
| | Medium | 24,046 | 69 | 14.71 | 29.73 | 22.08 | 12.40 | 8.10 | 19.56 | 44.9% |
| | High | 33,990 | 73 | 24.37 | 56.40 | 44.88 | 20.17 | 15.21 | 39.72 | 37.6% |
| Acceleration | Low | 23,198 | 71 | 11.51 | 26.89 | 22.42 | 10.78 | 7.54 | 20.21 | 34.4% |
| | Medium | 27,627 | 77 | 15.17 | 33.54 | 25.09 | 12.80 | 9.25 | 25.31 | 39.1% |
| | High | 31,721 | 73 | 21.77 | 55.23 | 25.19 | 15.51 | 10.84 | 36.98 | 50.2% |
| Turning | Low | 30,570 | 73 | 19.69 | 50.33 | 43.85 | 18.13 | 13.91 | 38.15 | 29.4% |
| | Medium | 24,432 | 72 | 13.16 | 31.98 | 22.59 | 11.80 | 8.49 | 25.29 | 35.5% |
| | High | 27,531 | 66 | 15.23 | 37.08 | 12.93 | 10.48 | 6.48 | 16.78 | 57.5% |

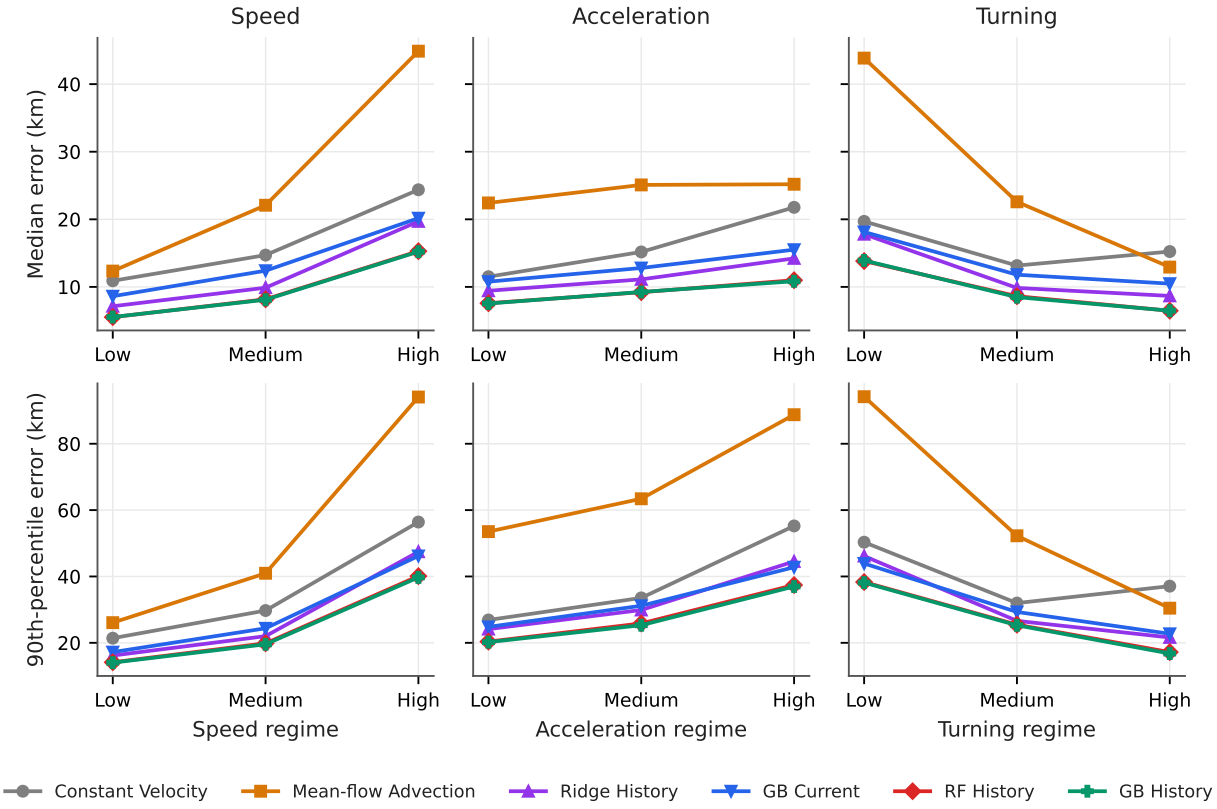



**Figure 6:** Median (top) and 90th-percentile (bottom) test error across speed, acceleration and turning regimes for six representative methods. Lines join regime summaries for readability only; the regimes are ordered categories. RF History and GB History almost coincide. Re-plotted from `models/motion_conditioned_test_errors.csv` (notebook cell 51).

**Speed.** Error rises steeply with recent speed for every method. For GB History the median roughly triples from 5.55 km (Low) to 15.21 km (High), and P90 rises from 14.0 km to 39.7 km (bottom row of Fig. 6). The P90 profiles mirror the medians for every descriptor, so the patterns are not confined to typical forecasts. The relative advantage over CV shrinks with speed (49% to 38%), but the absolute advantage grows (5.4 km, 6.6 km and 9.2 km). Part of this gradient is geometric: a fixed directional error produces a larger position error the farther the drifter travels. Even the best model therefore remains least accurate, in absolute terms, for the fast drifters that move material farthest.

**Acceleration.** Errors also increase with acceleration magnitude, but the pattern differs from speed. CV degrades fastest (median 11.5→21.8 km), while GB History degrades more slowly (7.5→10.8 km), so the relative gain rises from 34% to 50%. This is what one would expect if the history summaries capture changes in velocity that constant extrapolation ignores. The acceleration magnitude mixes changes in speed and direction, and for a given turning rate it scales with speed, so this descriptor is not independent of the other two.

**Turning.** Contrary to the proposal's working hypothesis, which expected turning to be associated with *larger* errors, errors are highest for the *least* turning drifters (GB History median 13.91 km) and lowest for the most turning ones (6.48 km). Two further observations help interpret this. First, Mean-flow Advection, which is far worse than CV overall, beats CV in the High-turning regime (12.93 against 15.23 km), and it is catastrophically wrong in the Low-turning regime (43.85 km). When a drifter is looping, its instantaneous velocity is a poor guide to its net 24-hour displacement. A climatology that predicts small net motion is then preferable, whereas in a straight, fast jet the climatology is far too slow. Second, the relative gain of GB History over CV is largest under high turning (57.5%), consistent with the history features summarising oscillatory or curved motion. These interpretations are plausible but untested. Because turning and speed are correlated, the turning pattern must be examined jointly with speed.

## 6.2 Joint speed–turning analysis

Figure 7a confirms a strong association between the two descriptors. Among Low-speed forecasts, 62.4% are High-turning, whereas 71.8% of High-speed forecasts are Low-turning, and the High-speed/High-turning cell contains only 3,816 forecasts (4.6%) from 44 drifters. Much of the marginal "high turning means low error" pattern therefore reflects the fact that strongly turning drifters are usually slow.

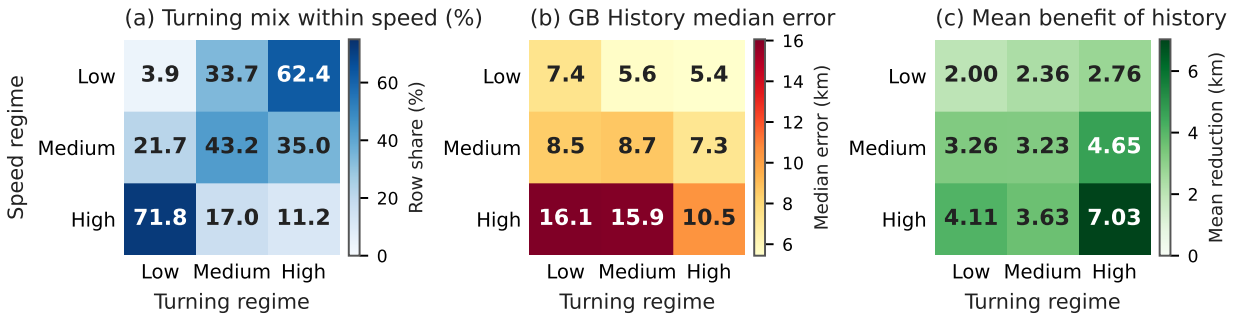



**Figure 7:** Joint speed–turning regimes on the test set (82,533 forecasts with defined turning). (a) Turning distribution within each speed regime (row %). (b) Median GB History error (km). (c) Mean paired benefit of history, $E^{\mathrm{GBC}} - E^{\mathrm{GBH}}$ (km); note that (b) shows medians and (c) shows means. Re-plotted from saved test predictions; values reproduce notebook cell 53.

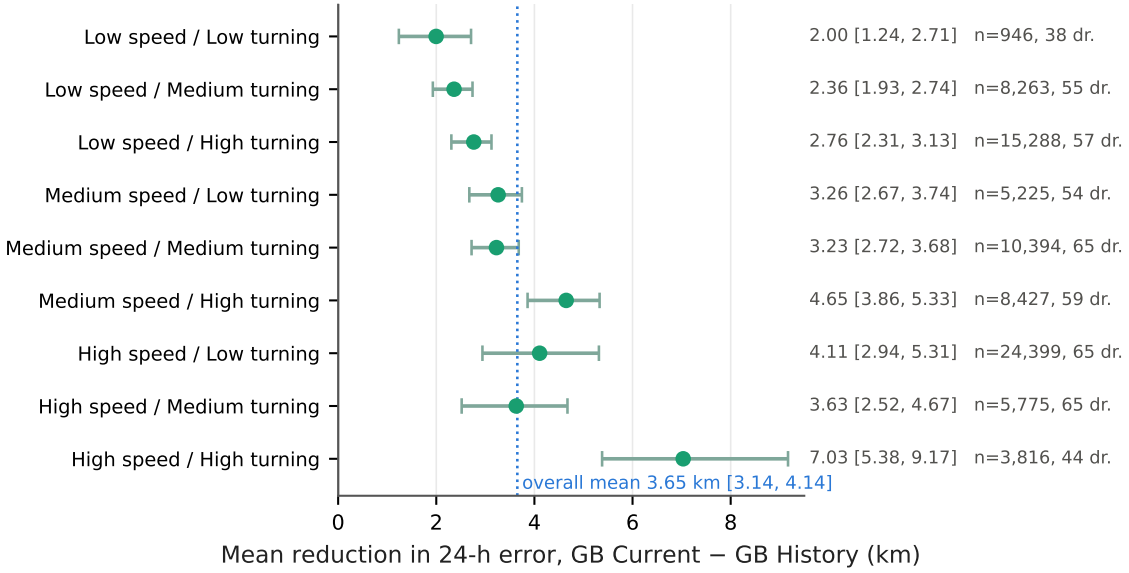
The association does not, however, explain the turning pattern away. Within each speed tertile, High-turning forecasts still have the lowest GB History median error (Fig. 7b): 16.06, 15.94 and 10.47 km across turning regimes at High speed, and 7.36, 5.57 and 5.44 km at Low speed. The Medium-speed row is not monotone (8.48, 8.68, 7.26 km). The tertiles are broad, and the High-speed band includes every mean speed above 0.40 m s$^{-1}$, so residual confounding by speed within a band cannot be excluded.

The mean benefit of history (Fig. 7c) increases with speed in every turning column and is largest in the High-turning column of every speed row. It ranges from 2.00 km (Low speed, Low turning) to 7.03 km (High speed,

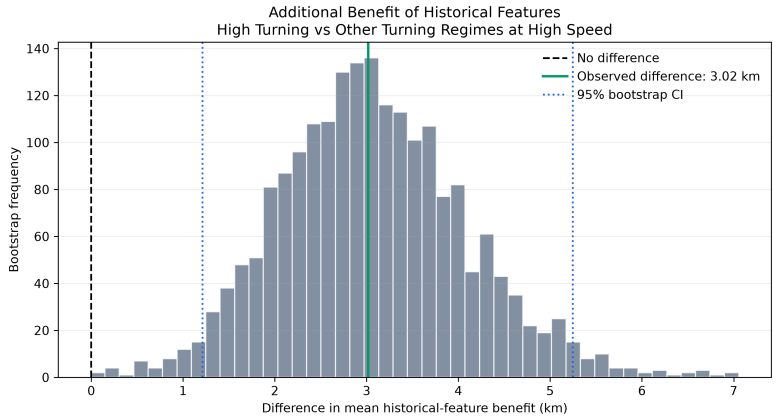
High turning). Part of the speed gradient is a scale effect, since larger errors leave more room for absolute improvement. The High-speed/High-turning cell is an exception to a pure scale explanation: its GB History median error (10.5 km) is *lower* than in the other High-speed cells, yet the benefit of history there is the largest.

### 6.3 Uncertainty in the benefit of history

Figure 8a gives drifter-cluster bootstrap intervals for the nine joint-regime benefits (Table 9 in Appendix A.1). All nine point estimates are positive and every 95% interval excludes zero. Even the smallest benefit, 2.00 km (1.24 km to 2.71 km) for Low speed and Low turning, is distinguishable from zero after accounting for within-drifter dependence. The High-speed/High-turning interval is the widest (5.38 km to 9.17 km), reflecting its 44 contributing drifters.




**(a)** Mean benefit of history by joint regime, with 95% intervals, cases and drifters




**(b)** High-speed contrast: High vs Low/Medium turning

**Figure 8:** Drifter-cluster bootstrap (2,000 resamples of the 78 test drifters). (a) Mean paired benefit of history ($E^{\mathrm{GBC}} - E^{\mathrm{GBH}}$) with 95% percentile intervals; the dotted line is the overall mean. Re-plotted from `models/history_gain_cluster_bootstrap.csv` (cell 55). (b) Bootstrap distribution of the difference in mean benefit between High-turning and Low/Medium-turning forecasts within the High-speed regime (original figure, cell 57).

Separate intervals for each regime do not by themselves test whether two regimes differ. We therefore estimated the contrast that motivated the analysis directly (Fig. 8b). Within the High-speed regime, the mean benefit was 7.03 km for High-turning forecasts and 4.02 km for Low/Medium-turning forecasts. The difference of 3.02 km has a 95% bootstrap interval of 1.21 km to 5.25 km, which excludes zero. This contrast was

defined *after* inspecting Fig. 7c, it is one of many comparisons that could have been made, and no multiplicity adjustment was applied. It should therefore be read as strong exploratory evidence that history is especially valuable for fast drifters that are turning, to be confirmed on independent data, rather than as a pre-registered test. The analysis also does not separate turning from co-varying factors such as location or the eddy field.

**Answer to the research question (motion).** In this dataset, 24-hour forecast difficulty is driven primarily by recent speed. Acceleration adds difficulty mainly for Constant Velocity. Turning is associated with *lower* absolute error, largely through its negative association with speed. History improves forecasts in all regimes examined, and most clearly for fast, turning motion.

# 7 Spatial, temporal and environmental context

## 7.1 Spatial variation

Test forecasts were aggregated by origin on a 1° grid, keeping cells with at least 100 forecasts from at least three distinct drifters. This retains 145 of 198 occupied cells and 73,214 forecasts (88.7%) (Fig. 9). Because the split was by drifter, not by location, test drifters pass through the same areas as training drifters. The maps therefore describe *where* the model is more or less accurate; they do not test generalisation to unseen regions.

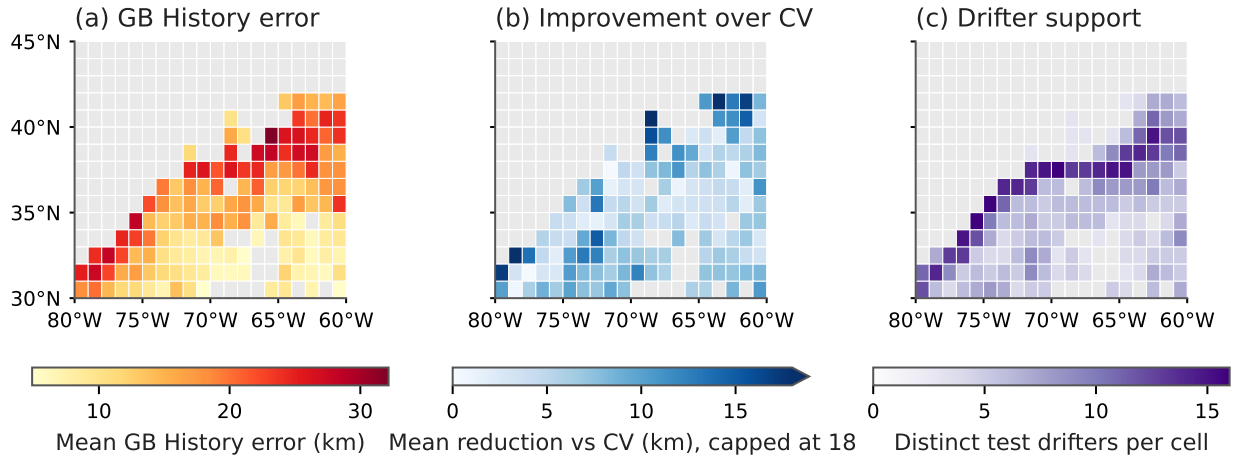



**Figure 9:** Spatial variation in held-out performance (cells with ≥100 forecasts and ≥3 drifters; grey: excluded or unsampled). (a) Mean GB History error. (b) Mean reduction in error relative to CV. The colour scale is capped at 18 km (98th percentile of cell values) for readability; three cells exceed it: 42.0 km at 40°N 69°W (3 drifters), 20.4 km at 32°N 79°W and 18.5 km at 41°N 64°W (cells labelled by south-west corner). No cell has a negative mean reduction. (c) Distinct test drifters per cell. Re-plotted from saved test predictions with the notebook's aggregation rules, which were verified to reproduce its 145 cells and 73,214 forecasts; the original figure used a symmetric ±42 km scale set by the single largest cell.

Absolute difficulty is spatially structured. Mean GB History errors exceed 25 km in a band running from the south-western corner of the box north-eastward to about 38°–40°N, 60°–68°W. The worst cell (39°N 66°W) averages 32.2 km but rests on only three drifters. Errors are below 10 km in much of the south-central box. The high-error band broadly follows the region where one would expect the Gulf Stream jet and its extension, and it coincides with the cells of highest drifter support (Fig. 9c). We did not overlay an independent current-axis product, however, so this correspondence remains qualitative.

The *improvement* over CV is positive in all 145 cells (median 6.2 km, minimum 0.3 km), so the advantage of the learned correction is not confined to a sub-region. High-error cells are not necessarily cells of low improvement. For example, the cell at 38°N 67°W combines a 27.3 km mean error with an 11.3 km improvement, whereas the cell at 39°N 64°W improves by only 1.9 km. Cell estimates are noisy, because most cells rest on 3–10 drifters (median 6) whose forecasts are strongly autocorrelated, and individual cells should not be over-interpreted.

## 7.2 Temporal stability

GB History had the lowest median error of the three compared methods (CV, GB Current, GB History) in every one of the 16 years, and GB Current beat CV in every year. Annual GB History medians ranged from 6.5 km (2013) to 16.0 km (2009), but each year contains only 3–13 test drifters, so year-to-year variation largely reflects which drifters happen to be present. Aggregated into four periods (Table 6), the advantage is remarkably stable. The mean reduction relative to CV is 7.1 km to 7.8 km and the mean benefit of history 3.4 km to 4.2 km in every period. Because training and test drifters cover the same years, this shows temporal consistency within 2007–2022, *not* prospective skill in future years.

**Table 6:** Test performance by period (km). Gains are mean paired reductions. Drifters may contribute to more than one period. Source: notebook cell 63; yearly values are plotted in Fig. 10.

| | | | Median error | | | Mean gain of GB History | |
|---|---|---|---|---|---|---|---|
| Period | Cases | Drifters | CV | GB Current | GB History | vs CV | vs GB Current |
| 2007–2010 | 18,316 | 22 | 15.63 | 13.57 | 9.46 | 7.12 | 4.20 |
| 2011–2014 | 27,654 | 30 | 15.42 | 12.40 | 8.77 | 7.29 | 3.36 |
| 2015–2018 | 20,056 | 14 | 16.92 | 13.67 | 10.10 | 7.12 | 3.48 |
| 2019–2022 | 16,520 | 15 | 15.71 | 12.84 | 8.85 | 7.82 | 3.73 |

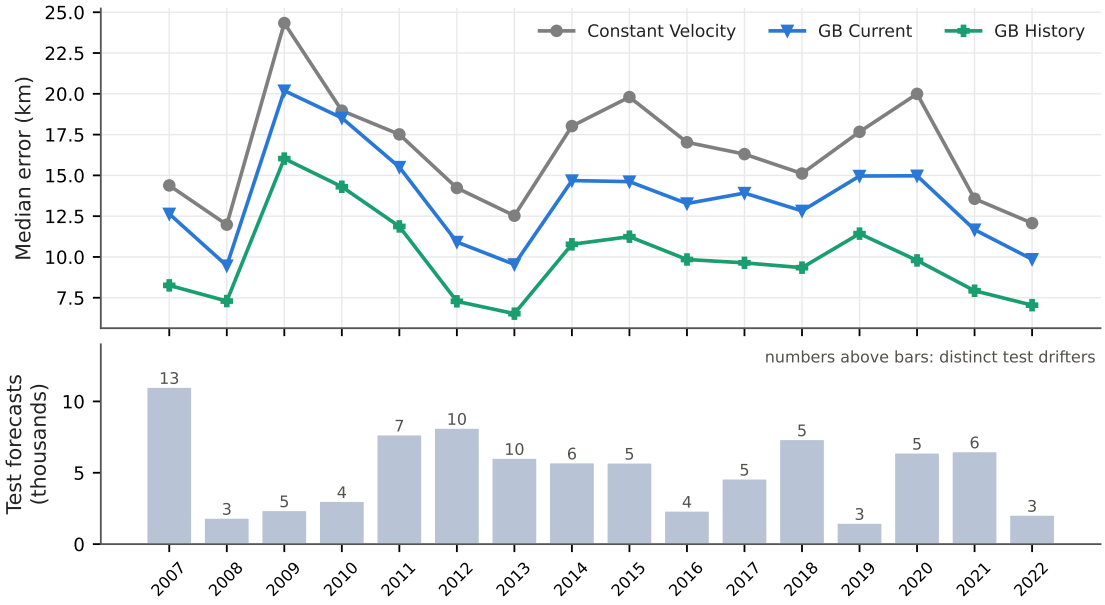



**Figure 10:** Median test error by year for CV, GB Current and GB History (top), and the number of test forecasts and distinct test drifters per year (bottom). Several years rest on three to five drifters. Re-plotted from saved test predictions; values reproduce notebook cell 63.

## 7.3 Sea-surface temperature: association versus predictive value

SST is an environmental variable rather than a kinematic one, so we examined it in two distinct ways.

**SST as a descriptive stratifier.** Using training-tertile thresholds (26.93 °C and 28.06 °C) on the 77,845 test forecasts with origin SST, errors decrease from the Low to the High SST regime for all three methods (Fig. 11a–c): GB History medians are 9.98 km, 9.28 km and 8.28 km and CV medians 17.59 km, 15.50 km and 14.93 km. The mean benefit of history is slightly larger at Low SST (4.09 km against 3.49 km and 3.39 km), although median benefits are almost identical (3.34 km to 3.47 km). SST is confounded with speed: High-speed forecasts make up 43.5% of the Low-SST regime and about 40% of the others, while Low-speed forecasts are most common at High SST (35.8% against 24.4%) (Fig. 11d). Within speed tertiles the SST gradient persists at low speed (6.44→4.81 km) and medium speed (8.61→7.57 km) but essentially vanishes at high speed (14.96 km to 15.77 km). SST is also tied to location and water mass (the Low regime has mean 24.2 °C

against 28.8 °C for High), so these patterns cannot be attributed to temperature itself.

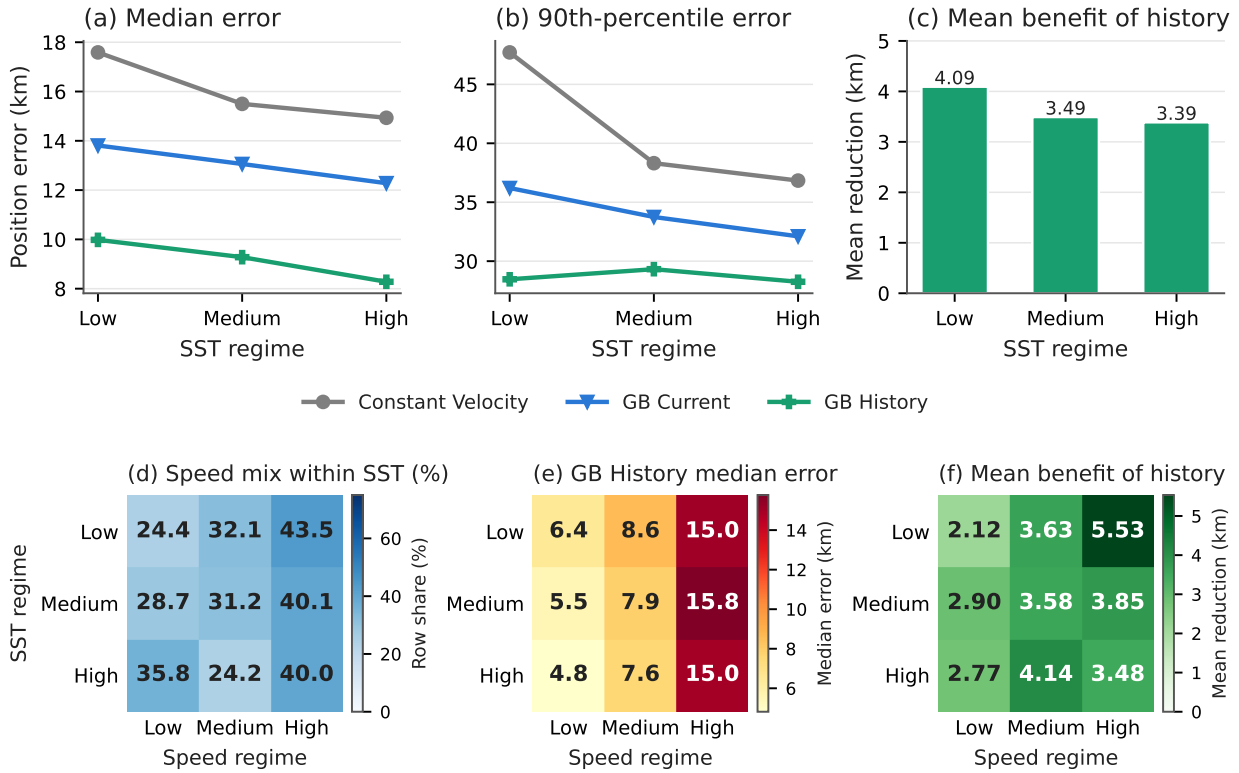



**Figure 11:** SST-conditioned test performance (77,845 forecasts with origin SST). (a–c) Median and P90 error and mean benefit of history by SST regime. (d–f) Joint SST–speed regimes: speed mix within each SST regime (row %), GB History median error, and mean benefit of history. Descriptive associations only. Re-plotted from saved outputs; values reproduce notebook cells 68 and 70.

**SST as an additional predictor.** To test whether SST adds information *beyond* kinematics, GB History was refitted with origin SST as a 26th feature, keeping the hyperparameters, data and targets identical, and evaluated on the validation set (Table 7). The changes are negligible and mixed: the mean falls by 0.019 km (0.16%) and P90 by 0.05 km, but the median rises by 0.03 km. Only 50.1% of individual forecasts improve, and the median paired change is 0.002 km. Under this configuration, a single instantaneous SST value carries no useful additional predictive information. One possible explanation is that what SST says about the flow regime is already captured by position, velocity and recent motion. A richer representation, such as SST gradients or fronts, might behave differently. Because this experiment was run after the test set had been examined, it is reported as exploratory validation evidence.

**Table 7:** SST feature ablation on the validation set (88,471 forecasts, km). Source: notebook cell 74.

| Model | Mean | Median | P90 | RMSE |
|---|---|---|---|---|
| GB History (25 features) | 11.969 | 8.507 | 24.977 | 16.681 |
| GB History + SST (26 features) | 11.950 | 8.537 | 24.929 | 16.633 |

# 8 Discussion

## 8.1 Synthesis

The results form a consistent picture. Constant Velocity is a strong baseline at 24 hours, far stronger than a climatological mean flow, because the instantaneous velocity already encodes the synoptic state of meanders and eddies. Its errors are nonetheless large and systematic, and they grow wherever velocity is not persistent: at high speed, under acceleration and during looping motion. A correction learned from the preceding 24

hours of the drifter's own motion removes about 40% of the median CV error on unseen drifters. Most of that gain comes from the history information rather than from model flexibility, but flexible tree ensembles exploit the history information more fully than linear models. These conclusions agree with the literature that strong kinematic baselines are hard to beat (Grossi et al., 2025; Lin et al., 2026). They also show that, at a 24-hour horizon and with hourly data, motion history carries substantial additional information, in line with Aksamit et al. (2020).

The motion analysis refines the proposal's working hypothesis. We expected turning and acceleration to make forecasts harder. Instead, speed dominates absolute difficulty, and turning is associated with *easier* forecasts, mostly because turning drifters are slow. Acceleration and turning matter more for *which method* works best than for absolute difficulty: CV is most disadvantaged under acceleration and turning, the climatology becomes competitive under strong turning, and history gives its largest benefit for fast, turning motion. Evaluating forecast methods only on aggregate error would miss all of these distinctions.

## 8.2 Practical significance of the findings

We separate what was demonstrated (retrospective skill on held-out GDP drifters) from what it could mean for applications (untested).

- **Size of the improvement.** The typical 24-hour miss shrinks from 15.89 km to 9.23 km (41.9%), and the 90th-percentile miss from 40.90 km to 28.55 km. As a purely geometric illustration, a circle with these radii would cover about one-third (median) and one-half (P90) of the original area. Real search or monitoring areas also depend on the object's own drift, the shape of the error distribution and operational factors, so this is not a demonstrated reduction in search effort.
- **Knowing in advance when a forecast is unreliable.** All three motion descriptors are computed from data available at forecast time. The GB History median error triples from low- to high-speed regimes (5.55 to 15.21 km) and the P90 rises to 39.7 km. Recent speed therefore acts as a usable, if coarse, warning that a particular forecast deserves wider uncertainty margins. A single aggregate error would hide this.
- **Where history matters most.** Among fast drifters, adding history reduced mean error by 7.03 km for high-turning forecasts versus 4.02 km for low/medium-turning forecasts. The between-group difference was 3.02 km (95% drifter-cluster bootstrap CI: 1.21·5.25 km). This post-hoc comparison is exploratory; rapid turning at high speed can undermine velocity persistence.
- **Curvature is not a reliable warning sign.** Strongly turning tracks had the *lowest* errors (GB History median 6.48 km against 13.91 km for nearly straight tracks), mainly because turning drifters are slow. A user who treated a curvy track as the main signal of an unreliable forecast would misdirect attention; speed is the better first indicator, and turning is more informative about *which* method to use.

## 8.3 Limitations

We separate limitations of the *study design*, which bound what the evidence can show, from *demonstrated weaknesses* of the models.

**Study design.**

(i) *Retrospective inputs.* GDP hourly positions and velocities are estimated from satellite fixes on both sides of each time (Elipot et al., 2016). The "current" velocity and the last hours of the input window therefore contain some information from after $t$. This affects every method, including the CV baseline, and the size of its effect on skill is unknown. The results measure retrospective predictability, not real-time operational skill.

(ii) *Drogue status.* 48.7% of forecast origins come from drifters flagged as undrogued, whose motion includes wind-driven slip. The models neither filtered nor conditioned on drogue status, so the reported skill describes a mixture of two kinds of drifter behaviour, and performance within each group is unknown. This check was made only at the reporting stage.

(iii) *One split, 78 test drifters.* All results come from a single drifter-disjoint partition. We quantified uncertainty with the cluster bootstrap only for the GB Current versus History comparison; model-versus-

baseline differences and differences among the tree ensembles were not bootstrapped, and the variability across alternative splits was not measured. The very large and consistent margins over CV make a reversal implausible, but the exact percentages are uncertain.

(iv) *Dependence and test reuse.* The cluster bootstrap handles dependence within drifters but not dependence between drifters sampling the same features. The subgroup, spatial, temporal and SST analyses reused the test set after the models were frozen, and the high-speed contrast was chosen after inspecting the data.

(v) *Fixed configuration.* There is one horizon (24 h), one history length (24 h), one season and one region. The course brief's suggested "error against lead time" comparison was not produced. Rolling forecasts re-initialised from observations say nothing directly about autonomous multi-step trajectory prediction.

(vi) *Tuning and loss.* The grids were small, the history tree models selected the most complex configuration available, and Gradient Boosting's default internal early stopping used a random row-level, not drifter-grouped, 10% of training rows, which may have stopped later than a grouped split would (Section 4); this stays inside the training drifters and does not touch the test set. The models were fitted with a squared-error loss in displacement but selected on median great-circle error. None of this compromises the test evaluation, but better-tuned models might perform somewhat better.

(vii) *Baseline and geometry.* The mean-flow field is an unweighted 1° average of training origin velocities. Drifters over-sample convergent regions, so such averages are biased estimates of the mean current, and a carefully constructed climatology (e.g. Lumpkin and Johnson, 2013) might do better. All geometry assumes a sphere with $R = 6{,}371$ km. The local-Cartesian versus spherical inconsistency was resolved (Section 3), but ellipsoidal effects, typically a fraction of a percent of distance, were not modelled.

(viii) *Observational associations.* Speed, acceleration, turning, SST and location are inter-correlated, and the models contain no wind, sea-surface-height or current-field information. Regime differences are associations and do not identify physical mechanisms.

**Demonstrated model weaknesses.** Even the best model retains a heavy upper tail (P90 28.5 km overall, 39.7 km at high speed). Adding history made 28.6% of individual test forecasts worse (GB History against GB Current). Consecutive rolling forecasts jitter, and the models produce single deterministic endpoints with no statement of their own uncertainty.

### 8.4 Evaluation against the proposal's success criteria

Table 8 evaluates the project against the six success criteria fixed in the proposal. Beyond the original plan, the project (a) added Elastic Net, Random Forest and Extra Trees, (b) replaced single-step with stepwise mean-flow advection, (c) corrected the target geometry, and (d) added spatial, temporal and SST analyses and an SST ablation. Several planned diagnostics were not completed: results by drogue status, month or data quality; a sensitivity analysis weighting drifters equally; a mean-flow velocity feature; heading-sector analyses; and bootstrap intervals for comparisons against the physical baselines. An earlier exploratory analysis for the poster used a larger box (80°–50°W, 25°–45°N), a different sample definition and partly different metrics. Its numbers are not comparable with, and are superseded by, the results reported here.

### 8.5 Recommendations

The recommendations are addressed to the industry partner and to other users of the GDP hourly product. They apply to *retrospective* evaluation of the product: how predictable drifter positions are from the data themselves, and under which conditions. They are *not* yet evidence of operational real-time forecasting skill, because the inputs are smoothed using later fixes, almost half the cohort is undrogued, and only one region, season and horizon were tested.

1. **Benchmark for retrospective evaluation.** When assessing the 24-hour predictability of GDP hourly trajectories, use a history-enhanced tree-ensemble correction to Constant Velocity as the reference method. Gradient Boosting matched the accuracy of Random Forest and Extra Trees at a fraction of their training cost and storage. Do not rely on an unweighted climatological mean flow, except possibly as a component for strongly looping motion.
2. **Report errors by regime, with drifter-level uncertainty.** Median error varied threefold with recent

**Table 8:** Proposal success criteria and their status.

| Criterion (abridged) | Status | Evidence or gap |
|---|---|---|
| 1. Valid samples with full 24-h history and exact 24-h target; exclusions documented | Met | 99.04% of origins retained (Section 2); exclusions counted but not broken down by reason |
| 2. Both physical baselines and learned models, with documented mean-flow fallback | Met | Section 3; fallback 0.73% of hourly steps (validation) |
| 3. Identical held-out cases; preprocessing and selection respect drifter partitions | Met | Training-only estimation; one test evaluation (Section 4) |
| 4. Value of history vs baselines and no-history models, median and P90, with dependence-aware uncertainty | Partly met | Point estimates for all comparisons; cluster-bootstrap intervals only for Current vs History (GB) |
| 5. Predefined motion-group analysis with counts, paired gains and uncertainty | Largely met | Training tertiles fixed before testing; counts and drifters reported; intervals for joint regimes; single-descriptor groups not bootstrapped |
| 6. Predicted vs actual positions shown; region changeable without rewriting | Largely met | Figs. 4 and 5; region set in a configuration cell, but some plotting cells hard-code the box extent |

speed, and method rankings changed with turning. Forecast evaluations of drifter data should report regime-conditioned medians and tails with drifter-cluster intervals rather than one aggregate score, and should treat high recent speed, not track curvature, as the first indicator of a low-confidence forecast.

3. **Before any operational claim**, re-evaluate with inputs that are available in real time (velocities derived only from fixes up to $t$), stratify by drogue status or include it as a feature, use repeated or chronological drifter hold-outs, and report error against lead time. Transfer to search-and-rescue or pollution applications would additionally require validation against the drift of the relevant objects (e.g. leeway or windage), not only of drogued drifters.
4. **Target the tail.** Future work should prioritise probabilistic or ensemble forecasts that can express large uncertainty for fast or accelerating drifters. It should also test richer environmental predictors, such as gridded currents, wind and SST fronts, rather than single point values, which added nothing here.

# 9 Conclusion

Using 626,679 forecast cases from 517 GDP drifters in the Gulf Stream, we evaluated two physical baselines and ten learned residual-correction models on 78 held-out drifters. Recent motion history substantially improves 24-hour position forecasts. History-enhanced Gradient Boosting reduced the median error of Constant Velocity from 15.9 km to 9.2 km, and history improved every model family, most strongly the nonlinear tree ensembles. Forecast difficulty is governed mainly by recent speed. Turning is associated with lower error because turning drifters tend to be slow, and the benefit of history is positive in every motion regime and largest for fast, turning drifters. The advantage is geographically widespread and stable across 2007–2022, and adding SST as a predictor did not help. Practically, the results show that a drifter's own recent track substantially narrows where it is likely to be a day later, and that recent speed indicates in advance which forecasts deserve least confidence. These conclusions concern retrospective forecasts from smoothed hourly data that mix drogued and undrogued drifters. Real-time inputs, drogue-aware evaluation and multiple lead times are the essential next steps toward an operational assessment.

# 10 Reflection on the use of AI tools

We used generative AI throughout the project. ChatGPT was used mainly during modelling, for Python code generation, debugging, alternative implementations, review of the geodesic calculations, visualisation design, evaluation and bootstrap code, and interpretation. Claude Code (Anthropic) was used during report preparation, to draft the report from the final notebook, write the figure-regeneration script and cross-check reported numbers against the saved outputs. All modelling decisions and every reported number were checked against the code and its actual outputs. Four concrete examples show where AI helped and where it was wrong.

**1. Trajectory visualisation (helped after iteration).** The first AI-suggested design overlaid the observed track and all model predictions on one map. With several models the lines were indistinguishable. Over several iterations we moved to one panel per model, fixed geographic extents per drifter so that panels are comparable, solid observed and dashed predicted lines, and in-panel window medians (Fig. 4). The final design came from judging the plots ourselves, not from the first suggestion.

**2. Geodesic consistency (AI code was wrong).** AI-assisted code computed observed displacements with a local Cartesian approximation but mapped predictions back with the spherical forward geodesic, mixing two conventions without flagging it. We identified the inconsistency, quantified it (P99 round-trip error 1.54 km, maximum 2.98 km) and replaced the targets with an inverse-geodesic formulation before refitting all models (Section 3). The lesson is that generated code can be internally plausible yet mathematically inconsistent.

**3. Methodological wording (needed correction).** AI-generated descriptions did not reliably distinguish our single drifter-disjoint hold-out from $K$-fold cross-validation, or rolling 24-hour forecasts re-initialised from observations from recursive trajectory forecasting. We checked the wording against the code, which uses `GroupShuffleSplit` and no $K$-fold routine, and corrected it, because these distinctions determine what the results can claim.

**4. Report preparation (helped, with one error caught).** When drafting the bibliography, the assistant proposed a DOI from memory for a Gulf Stream reference. It resolved to an unrelated article and was discarded once every DOI was checked against Crossref. On the positive side, systematic cross-checking surfaced issues we had not documented: the undrogued share of the cohort, a shared legend in an existing figure whose error values apply to only one panel, and the default early-stopping behaviour of the boosting estimator. We verified each against the data or library documentation before including it.

Overall, AI accelerated coding, plotting and drafting. It was least reliable where correctness depended on domain conventions or precise experimental definitions, and human verification against outputs remained essential.

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# A Supplementary material

## A.1 Bootstrap results by joint regime

**Table 9:** Mean paired benefit of history ($E^{\mathrm{GBC}} - E^{\mathrm{GBH}}$, km) by joint speed–turning regime with 95% drifter-cluster bootstrap intervals ($B = 2000$). Source: `models/history_gain_cluster_bootstrap.csv` (notebook cell 55).

| Speed | Turning | Cases | Drifters | Mean benefit | 95% interval |
|---|---|---|---|---|---|
| Low | Low | 946 | 38 | 2.00 | [1.24, 2.71] |
| Low | Medium | 8,263 | 55 | 2.36 | [1.93, 2.74] |
| Low | High | 15,288 | 57 | 2.76 | [2.31, 3.13] |
| Medium | Low | 5,225 | 54 | 3.26 | [2.67, 3.74] |
| Medium | Medium | 10,394 | 65 | 3.23 | [2.72, 3.68] |
| Medium | High | 8,427 | 59 | 4.65 | [3.86, 5.33] |
| High | Low | 24,399 | 65 | 4.11 | [2.94, 5.31] |
| High | Medium | 5,775 | 65 | 3.63 | [2.52, 4.67] |
| High | High | 3,816 | 44 | 7.03 | [5.38, 9.17] |
| All (defined turning) | | 82,533 | 78 | 3.65 | [3.14, 4.14] |

## A.2 Reproducibility map

All results come from `notebook/08_reoprt_part_1.ipynb` (sic), executed top to bottom. The main artefacts are listed below; the report's own figure script is `report/scripts/make_figures.py`, which reads only saved outputs.

| Result | Notebook cell(s) and saved artefact |
|---|---|
| Cohort, features, split (Tables 1 and 2) | cells 4–8; `data/processed/modelling_dataset_final.parquet` |
| Baselines, geometric check | cells 9–10, 19, 23, 25 |
| Tuning and validation (Table 3) | cells 27–36; `models/*_validation_results.csv`, `final_validation_comparison.csv` |
| Test evaluation (Table 4, Fig. 3) | cell 39; `models/final_test_comparison.csv`, `data/processed/report_test_predictions.parquet` |
| Motion regimes, joint, bootstrap (Table 5, Figs. 6 to 8) | cells 49–57; `models/motion_*.csv`, `history_gain_cluster_bootstrap.csv` |
| Spatial, temporal, SST (Figs. 9 to 11, Tables 6 and 7) | cells 60–74; `data/processed/report_sst_ablation_validation.csv` |
| Figures 1, 3, 9; drogue share | `report/scripts/make_figures.py` (saved outputs only) |

## A.3 Additional trajectory comparisons

Figures 12 and 13 extend Fig. 4 to the remaining baseline and history-enhanced models.

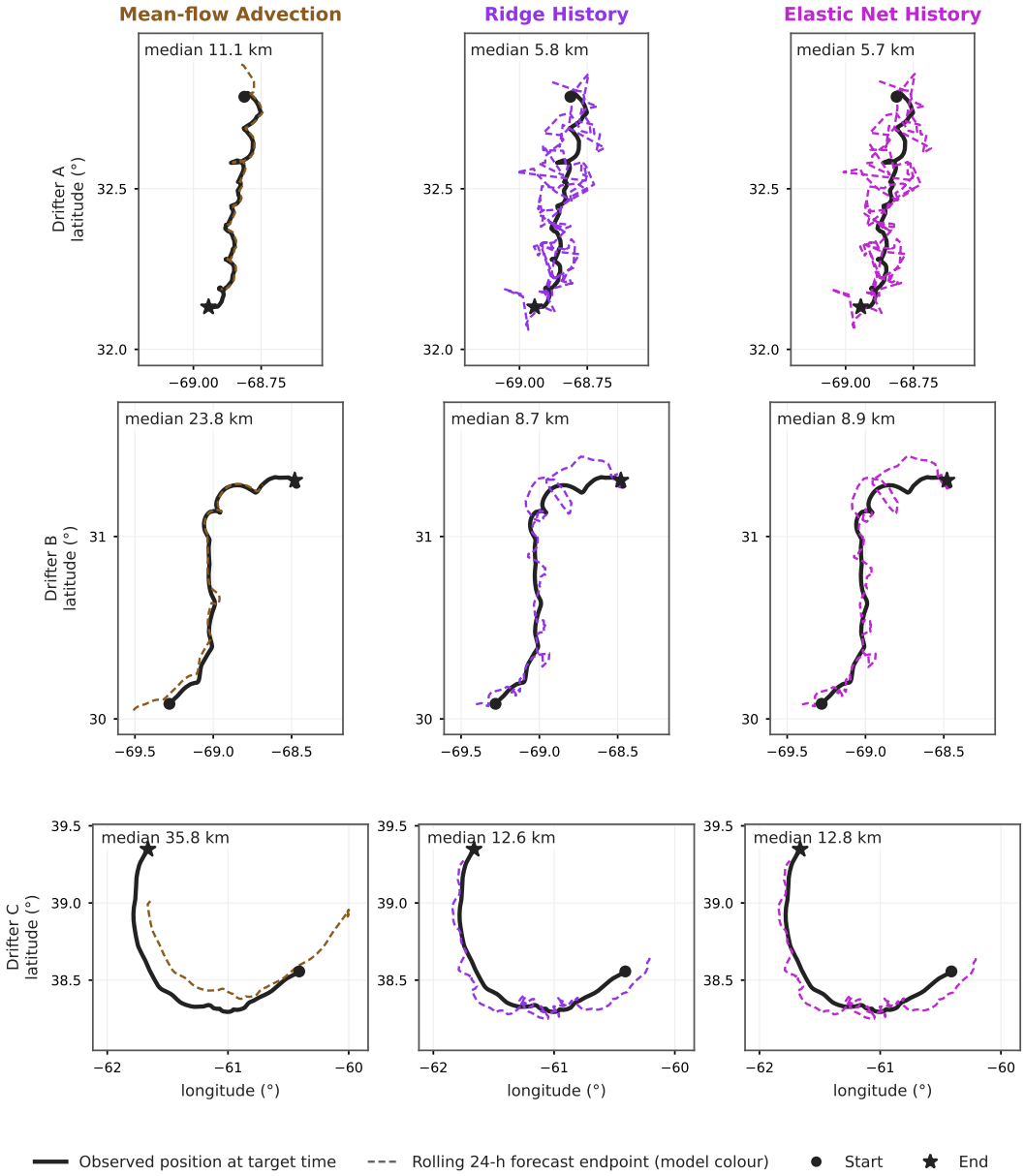



**Figure 12:** Rolling 24-hour forecast endpoints for Mean-flow Advection, Ridge History and Elastic Net History, for the same drifters, windows and axes as Fig. 4. Ridge and Elastic Net tracks are nearly identical; Mean-flow Advection has much larger errors for drifters B and C (window medians 23.8 and 35.8 km) than for A (11.1 km). Re-plotted from saved test predictions (notebook cell 46 selection).

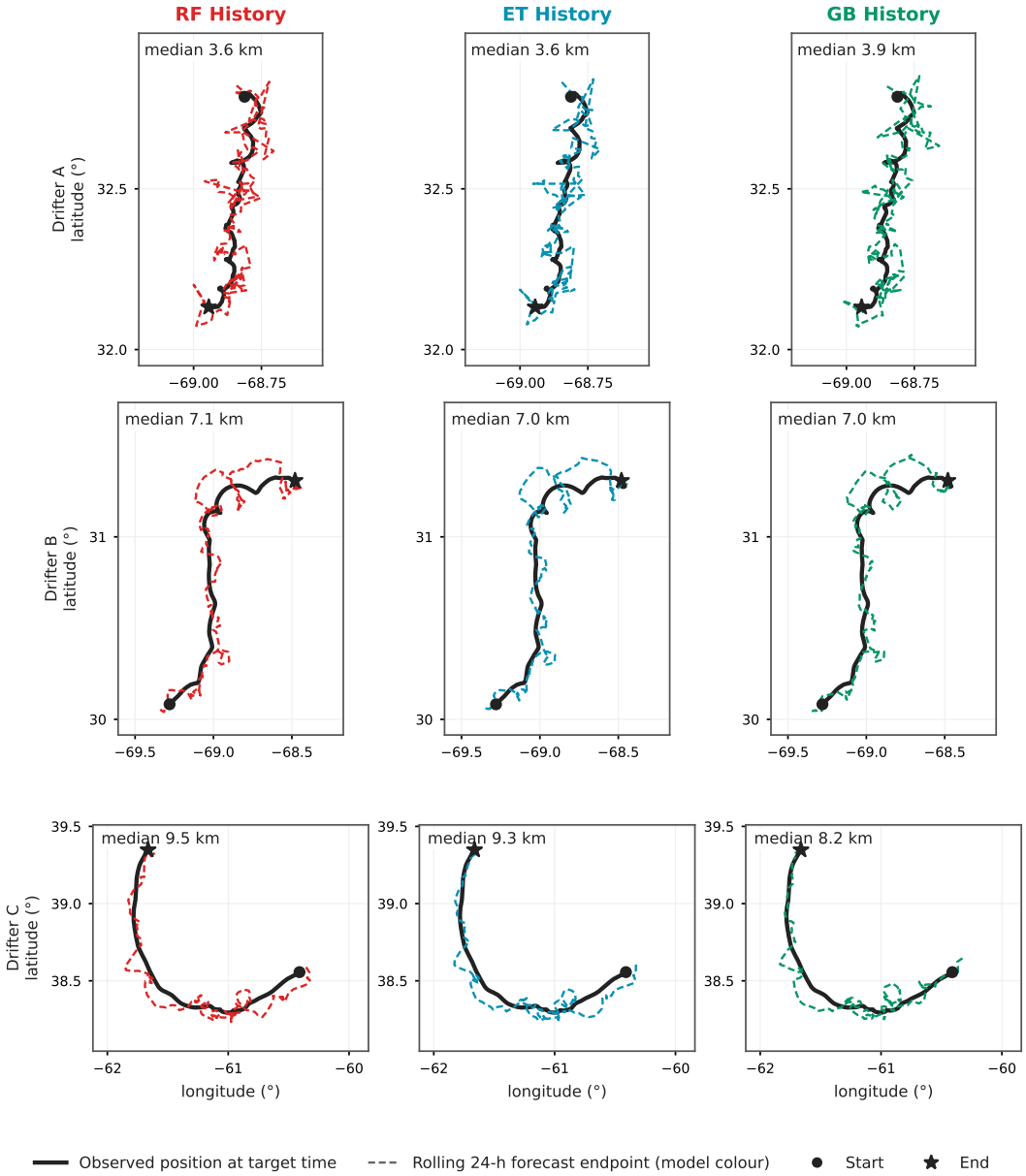



**Figure 13:** As Fig. 12 for RF History, ET History and GB History. The three tree ensembles produce visually indistinguishable rolling forecasts, consistent with their near-identical test errors (Table 4).